

Notes: Alonso, Dessein, and Matouschek (AER, 2008)

When Does Coordination Require Centralization?

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1 Big picture

- **Question.** The paper asks when efficient coordination inside a multidivisional organization requires centralization, and when decentralized managers can coordinate by communicating directly with one another.
- **Central trade-off.** Each division decision should adapt to its local state, but the two decisions should also be coordinated. Adaptation favors giving authority to the locally informed manager; coordination seems to favor headquarters.
- **Information problem.** Division managers privately observe their own local conditions. They cannot commit to truth-telling and communicate by cheap talk.
- **Incentive problem.** Division managers are biased toward their own division's payoff rather than total firm payoff. Headquarters is unbiased but must rely on vertical communication.
- **Organizational choices.** Under **centralization**, both decision rights are held by headquarters. Under **decentralization**, each division manager controls his own decision and communicates horizontally with the other manager.
- **Main surprise.** A higher need for coordination improves horizontal communication but worsens vertical communication. Therefore, decentralization can dominate centralization even when coordination is very important.
- **Why this is counterintuitive.** A naive view says that the more important coordination is, the more authority should be centralized. The paper shows this misses the endogenous communication response: when divisions know that coordination matters, they communicate more precisely with each other.
- **Core mechanism.** Vertical communication becomes harder when coordination is more important because managers know headquarters will impose a compromise. Each manager therefore has a stronger incentive to distort local information toward his division's ideal decision. Horizontal communication becomes easier because each manager internalizes that the other manager will make his own decision and coordination failure is costly for both.
- **Delegation principle.** The model formalizes a modern version of the local-knowledge argument for delegation. Local information can justify decentralization even without physical communication costs.
- **Economic takeaway.** Coordination does not mechanically imply centralization. The right organizational structure depends jointly on the importance of coordination and the degree to which division managers' incentives are aligned with firm-wide objectives.

2 Incremental contribution

- **From authority to communication quality.** Earlier theories often compare centralization and decentralization by assuming differences in authority, effort, or communication technology. This paper makes the *quality of communication* endogenous to the organizational structure.
- **Horizontal versus vertical cheap talk.** The paper distinguishes horizontal communication between biased division managers from vertical communication from biased division managers to unbiased headquarters.
- **No-commitment organizational design.** The organization can commit only to the allocation of decision rights, not to incentive-compatible mechanisms, transfers, or binding communication protocols. This makes the Revelation Principle inapplicable and gives decentralization a genuine role.
- **Coordination can help decentralization.** The key incremental insight is that a greater need for coordination does not simply increase the value of a central coordinator. It also changes communication incentives, making decentralized horizontal communication more informative.

- **Continuous coordination-adaptation trade-off.** The model allows decisions to be more or less coordinated and more or less adapted, rather than treating coordination as a binary outcome.
- **Formalizes the Delegation Principle under strategic communication.** The standard local-knowledge argument often assumes communication constraints. This paper shows that the argument survives even when physical communication is costless, because strategic misrepresentation remains.
- **Empirical implications.** The paper generates testable predictions: firms should centralize more when division managers have stronger own-division incentives and when coordination needs are high; the relationship should be weak when either incentive conflict or coordination need is low.
- **Extensions.** The paper shows that the central insight survives under sequential decision making and asymmetric divisions, though in weaker form.

3 Model objects and measurement

3.1 Organizational environment

The organization has two divisions and two decisions, d_1 and d_2 . Each decision should be close to its own local state, θ_1 or θ_2 , but the two decisions should also be close to each other.

- θ_j is the local state of Division j .
- Manager j observes θ_j privately.
- Headquarters observes neither state directly.
- Decisions are chosen after the managers communicate.

Interpretation: The model captures a common organizational problem: local units know local conditions, but firm value depends on both local adaptation and cross-unit coordination.

3.2 Profits and coordination

The paper writes divisional payoffs so that a manager values adaptation to his own state and coordination with the other division. A convenient way to read the payoff is:

$$\pi_j = -(d_j - \theta_j)^2 - \delta (d_j - d_k)^2, \quad j, k \in \{1, 2\}, j \neq k.$$

- The first term is the adaptation loss: the decision differs from the local state.
- The second term is the coordination loss: the two decisions differ from each other.
- $\delta \geq 0$ measures the need for coordination.

Interpretation: Higher δ makes coordination more valuable relative to adaptation. The central question is how authority allocation changes communication and therefore the final trade-off between the two losses.

3.3 Managerial bias

Division managers care about their own division more than the other division. The paper captures this with an own-division bias parameter $\lambda \in [1/2, 1]$.

- $\lambda = 1/2$ means the manager is perfectly aligned with firm-wide profits.
- Higher λ means the manager places more weight on his own division.
- A higher λ creates stronger incentives to distort information.

Interpretation: The same communication channel becomes less informative when sender preferences diverge more from receiver preferences.

3.4 Communication quality

The model measures communication quality by the residual variance of a local state after messages are observed:

$$E \left[(\theta_j - E[\theta_j \mid m_j])^2 \right] = S_l \sigma_j^2, \quad l \in \{C, D\}.$$

- S_C is the residual-variance term under centralization.
- S_D is the residual-variance term under decentralization.
- Lower S_l means better communication.
- Perfect communication corresponds to $S_l = 0$.

Interpretation: Communication quality is not exogenous. It is determined by the sender's incentive to exaggerate local information and the receiver's expected use of that information.

3.5 Organizational structures

- **Centralization:** headquarters receives vertical messages (m_1, m_2) and chooses both decisions (d_1, d_2) .
- **Decentralization:** the division managers exchange horizontal messages and each manager chooses his own decision.
- **Formal mechanism:** only the allocation of decision rights is contractible ex ante.

Interpretation: Centralization has a natural coordination advantage because one decision maker chooses both decisions. Decentralization has a natural adaptation advantage because the decision maker directly observes the local state.

4 Models and theoretical specifications

4.1 Timeline

The game has four stages:

1. Decision rights are allocated.
2. Managers observe local states θ_1 and θ_2 .
3. Managers send cheap-talk messages.
4. Decisions d_1 and d_2 are made.

Interpretation: Since communication is cheap talk and decision makers cannot commit to decision rules before messages arrive, the equilibrium must discipline information transmission through incentives rather than through contracts.

4.2 Decision making under centralization

Under centralization, headquarters maximizes expected total profits using posterior beliefs about θ_1 and θ_2 . Headquarters chooses decisions that are convex combinations of the posterior means:

$$\begin{aligned} d_1^C &= g_C E[\theta_1 | m] + (1 - g_C) E[\theta_2 | m], \\ d_2^C &= g_C E[\theta_2 | m] + (1 - g_C) E[\theta_1 | m]. \end{aligned}$$

The weight g_C depends on δ .

Interpretation: Headquarters is unbiased, so its decision rule is efficient given its information. The inefficiency under centralization comes from the loss and distortion of information in vertical communication.

4.3 Decision making under decentralization

Under decentralization, Manager j directly knows θ_j but has imperfect information about θ_k . His decision is a weighted average of his own state and his belief about the other manager's state:

$$d_j^D = g_D \theta_j + (1 - g_D) E[\theta_k | m_k], \quad j \neq k.$$

Interpretation: Decentralized managers adapt well to their own state but do not fully internalize the coordination externality. Their decisions are therefore too local relative to the first best.

4.4 Communication equilibria

The communication equilibrium partitions the state space into intervals. A sender reports the interval in which his state lies. The receiver updates beliefs and chooses actions accordingly.

- More intervals mean more informative communication.
- In the efficient equilibrium, the number of intervals goes to infinity.
- Interval sizes become very small near $\theta_j = 0$ and grow as $|\theta_j|$ increases.

Interpretation: Unlike Crawford-Sobel cheap talk models with a finite maximum number of intervals, this model permits arbitrarily fine communication because sender and receiver preferences are aligned at $\theta_j = 0$.

4.5 Residual variance formula

In the most efficient equilibrium, residual variance is

$$E \left[(\theta_j - E[\theta_j | m_j])^2 \right] = S_l \sigma_j^2, \quad S_l = \frac{b_l}{3 + 4b_l}.$$

Here b_l summarizes incentives to misrepresent information under structure $l \in \{C, D\}$.

Interpretation: The formula connects communication quality directly to incentive conflict. When $b_l = 0$, communication is perfect. As $b_l \rightarrow \infty$, residual variance approaches $1/4$.

4.6 Quality of vertical versus horizontal communication

Proposition 3 states:

- $S_C = S_D = 0$ when $\lambda = 1/2$.
- $S_D > S_C$ otherwise: vertical communication is generally more informative.
- $\partial S_D / \partial \lambda > \partial S_C / \partial \lambda > 0$.
- $\partial S_C / \partial \delta > 0 > \partial S_D / \partial \delta$, and S_C and S_D converge as $\delta \rightarrow \infty$.

Interpretation: This is the mechanism at the heart of the paper. Stronger own-division bias worsens both communication modes, but especially horizontal communication. Stronger coordination worsens vertical communication while improving horizontal communication.

4.7 Expected organizational performance

Expected profits can be written as:

$$\begin{aligned}\Pi_C &= -(A_C + (1 - A_C)S_C)(\sigma_1^2 + \sigma_2^2), \\ \Pi_D &= -(A_D + B_DS_D)(\sigma_1^2 + \sigma_2^2).\end{aligned}$$

The constants A_C, A_D, B_D depend on δ and λ .

Interpretation: Centralization loses because headquarters is imperfectly informed. Decentralization loses because managers are biased and imperfectly informed about each other. The comparison depends on how those losses vary with δ and λ .

4.8 Adaptation and coordination losses

Let AL_l be adaptation losses and CL_l be coordination losses under structure l . Lemma 2 states:

$$AL_C \geq AL_D, \quad CL_C \leq CL_D.$$

Therefore decentralization has an adaptation advantage and centralization has a coordination advantage. Relative performance is:

$$\Pi_C - \Pi_D = -\Delta AL + 2\delta \Delta CL.$$

Interpretation: The intuitive trade-off is formal: centralization coordinates better, decentralization adapts better. The key is that both advantages are endogenous because communication changes with organizational design.

4.9 Centralization versus decentralization

Proposition 5 states:

- If λ is sufficiently close to $1/2$, decentralization strictly dominates centralization for any $\delta > 0$.
- If δ is sufficiently small, decentralization strictly dominates centralization.
- If both δ and λ are large, centralization can dominate.

Interpretation: Centralization is optimal only when both the need for coordination and the own-division bias are high enough. If managers are sufficiently aligned, decentralization can dominate even when coordination is extremely important.

5 Theoretical design and empirical implications

5.1 Why the model avoids the Revelation Principle

The organization cannot commit to a mechanism that maps messages into decisions. Communication is cheap talk and the decision maker chooses actions after receiving messages.

Interpretation: This assumption is critical. If the organization could commit to an incentive-compatible mechanism, centralization would be weakly optimal by the Revelation Principle. Without commitment, the allocation of authority changes communication incentives.

5.2 Why coordination can favor decentralization

A high need for coordination changes the sender's incentives differently under vertical and horizontal communication.

- Under centralization, each division manager wants headquarters to tilt the common compromise toward his division.
- Under decentralization, each manager knows that coordination failure hurts both managers, so it becomes more valuable to communicate accurately.
- As δ rises, horizontal communication improves and vertical communication deteriorates.

Interpretation: The result is not that decentralization coordinates better mechanically. It is that decentralization can induce better information transmission when coordination becomes very important.

5.3 Cross-sectional predictions

The model predicts:

- Holding coordination need fixed, firms should be more centralized when division managers' incentives are more divisional and less firm-wide.
- Holding own-division bias fixed, firms should be more centralized when coordination needs are higher.
- These correlations should appear mainly when both incentive conflict and coordination needs are sufficiently high.
- Firms with well-balanced division-manager incentives should often decentralize even when coordination matters.

Interpretation: The model suggests that empirical work should measure not only coordination needs but also the incentive alignment of local managers.

5.4 Panel-data implications

If a technological shock increases the scope for interdivisional coordination, the probability of reorganization toward centralization should rise more for firms in which division managers' compensation is more divisional.

Interpretation: This creates a difference-in-differences style empirical implication: coordination shocks should have heterogeneous organizational responses depending on incentive alignment.

5.5 Limitations

- The model is theoretical and does not test its predictions econometrically.
- It abstracts from effort provision, formal incentive contracts, internal politics, repeated interactions, and multi-level hierarchies.
- The baseline is symmetric: two divisions, equal size, equal coordination needs, simultaneous decisions.
- Extensions show robustness but also indicate that asymmetries can make centralization more attractive.

Interpretation: The model is best read as a clean mechanism showing why coordination and centralization are not the same thing.

6 Section-by-section study guide

- **Introduction.** Sets up the debate between Chandler-style centralization and Sloan-style decentralized coordination.
- **Related literature.** Positions the paper between property-rights theory, team theory, mechanism design, and cheap-talk approaches.
- **Model setup.** Defines the two-division organization, the adaptation-coordination objective, managerial bias, and no commitment.
- **Decision making.** Characterizes optimal decisions given posterior beliefs.
- **Communication.** Solves for the efficient cheap-talk equilibrium and derives residual variance.
- **Centralization versus decentralization.** Compares adaptation and coordination losses and derives Proposition 5.
- **Empirical implications.** Translates the theory into cross-sectional and panel predictions for organizational form.
- **Extensions.** Shows how sequential decision making and asymmetries change but do not eliminate the main insight.
- **Conclusion.** Emphasizes the strategic, not physical, nature of communication costs.

7 Figures and theoretical result blocks

The paper has eight main theoretical figures and no empirical tables. Close figures are placed on the same row, as requested.

7.1 Figure 1: Timeline + Figure 2: Communication Equilibria

Read:

- Figure 1 shows the four-stage timing: allocation of decision rights, observation of local states, cheap-talk messages, and final decisions.
- Under centralization, messages travel vertically to headquarters; under decentralization, messages travel horizontally between division managers.
- Figure 2 shows that communication equilibria partition the state space into intervals.
- The intervals are finer near the center of the state space and coarser farther away.
- Messages do not reveal the exact state; they reveal the interval in which the state lies.

Detailed interpretation: Figure 1 matters because the allocation of authority occurs before information is realized. This makes organizational form a commitment device. Figure 2 matters because communication is not assumed to be either perfect or absent; it is endogenous and partially informative.

Economic message: Organizational design shapes the information that will be transmitted later. The key object is not simply who decides, but how authority affects cheap-talk informativeness.

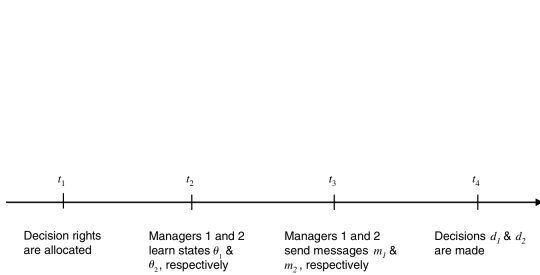


FIGURE 1. TIMELINE

Figure 1: Timeline

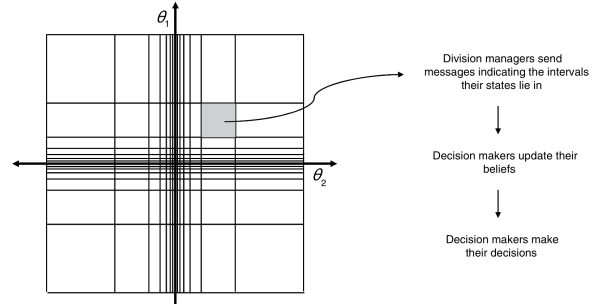


Figure 2: Communication Equilibria

7.2 Figure 3: Communication Comparison (λ) + Figure 4: Communication Comparison (δ)

Read:

- Figure 3 plots residual variance under decentralization, S_D , and centralization, S_C , as own-division bias λ rises.
- Both residual variances rise with λ , meaning communication worsens as managers become more biased.
- $S_D > S_C$: horizontal communication is less informative than vertical communication for a given $\lambda > 1/2$.
- Figure 4 plots residual variance as the need for coordination δ rises.
- S_C rises with δ , while S_D falls with δ .
- The two curves converge when δ becomes very large.

Detailed interpretation: Figure 3 shows the standard cheap-talk force: preference divergence reduces communication. Figure 4 delivers the paper’s distinctive result: greater coordination need can improve horizontal communication because decentralized managers care more about learning each other’s decisions.

Economic message: High coordination needs do not automatically favor centralization. They improve the incentives for decentralized managers to communicate with each other.

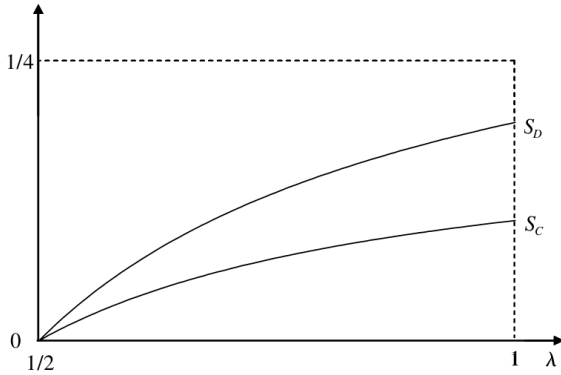


FIGURE 3. COMMUNICATION COMPARISON (λ)

Figure 3: Communication Comparison (λ)

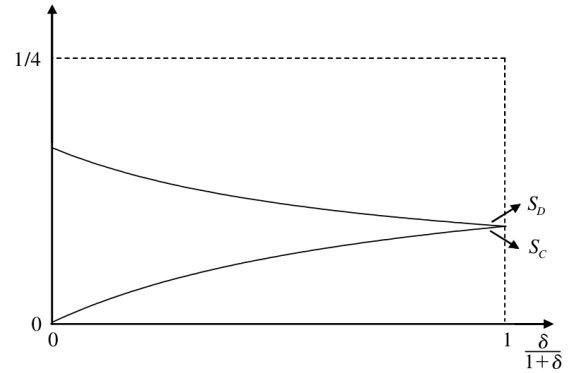


FIGURE 4. COMMUNICATION COMPARISON (δ)

Figure 4: Communication Comparison (δ)

7.3 Figure 5: Organizational Performance in Three Dimensions + Figure 6: Organizational Performance in Two Dimensions

Read:

- Figure 5 plots $\Pi_C - \Pi_D$ over λ and $\delta/(1 + \delta)$.
- Negative values mean decentralization is better; positive values mean centralization is better.
- Decentralization dominates when λ is close to $1/2$ even if δ is high.
- Centralization appears only when both own-division bias and coordination need are high.
- Figure 6 summarizes the regions in the (λ, δ) plane where each structure dominates.
- The boundary is downward sloping: for more biased managers, a lower coordination need can justify centralization.

Detailed interpretation: Figure 5 visualizes the full relative-performance surface. Figure 6 distills the main theorem. The vertical line at $17/28$ highlights the region where decentralization dominates for all coordination needs.

Economic message: The optimal organizational form is a complementarity between two forces: incentive misalignment and coordination need. Centralization requires both.

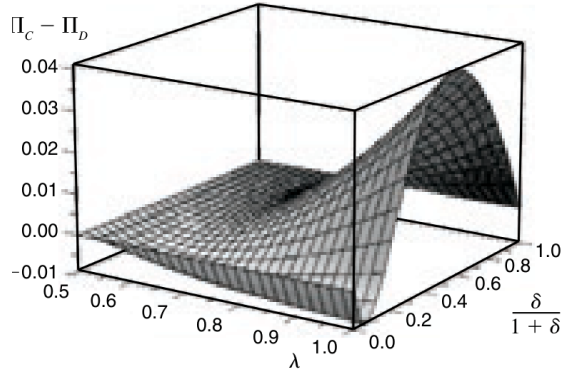


FIGURE 5. ORGANIZATIONAL PERFORMANCE
IN THREE DIMENSIONS

Figure 5: Organizational Performance in Three Dimensions

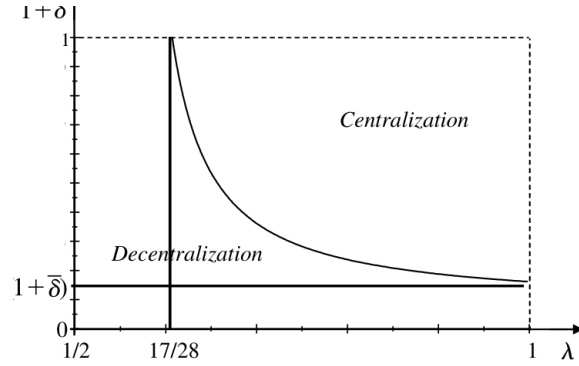


FIGURE 6. ORGANIZATIONAL PERFORMANCE
IN TWO DIMENSIONS

Figure 6: Organizational Performance in Two Dimensions

7.4 Figure 7: Sequential Decision Making with Equal Information + Figure 8: Sequential Decision Making with Unequal Information

Read:

- Figure 7 studies decentralization with sequential rather than simultaneous decisions and equal uncertainty in the two divisions.
- Sequential decision making weakens decentralization when coordination is very important.
- Still, decentralization dominates when own-division bias is low enough.
- Figure 8 allows the leader to have more private information than the follower.
- As the leader's relative information share $R = \sigma_1^2 / (\sigma_1^2 + \sigma_2^2)$ rises, decentralization becomes more attractive.
- This is because the leader's own information is perfectly used under decentralized authority.

Detailed interpretation: The extension clarifies that the baseline result is strongest under simultaneous symmetric decisions. Sequential decision making can favor centralization because the leader may over-adapt to his own state and the follower has a stronger incentive to distort information. But local knowledge still supports decentralization when incentive bias is small or the leader is especially well informed.

Economic message: Asymmetries can move the boundary, but the paper's main economic lesson remains: information advantages and incentive alignment can support decentralization even under strong coordination needs.

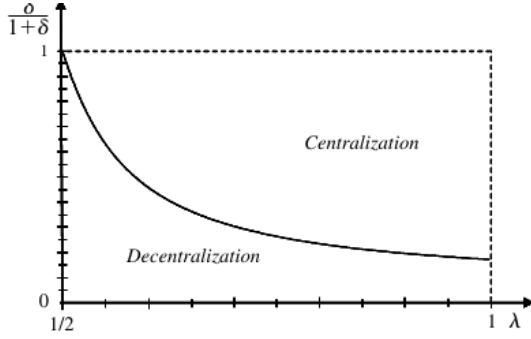


FIGURE 7: ORGANIZATIONAL PERFORMANCE WITH SEQUENTIAL DECISION MAKING WHEN $\sigma_1^2 = \sigma_2^2$

Figure 7: Sequential Decision Making, Equal Information

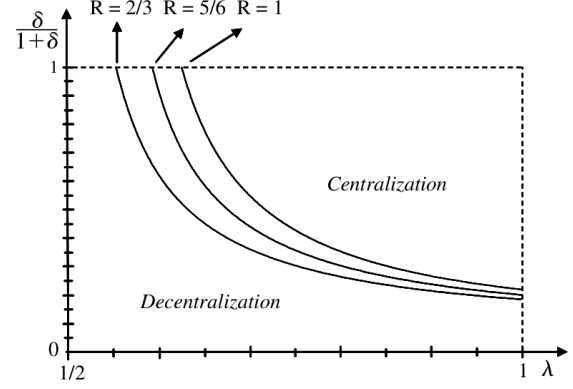


FIGURE 8: ORGANIZATIONAL PERFORMANCE WITH SEQUENTIAL DECISION MAKING WHEN $R \equiv \sigma_1^2 / (\sigma_1^2 + \sigma_2^2) > 1/2$

Figure 8: Sequential Decision Making, Unequal Information

8 Detailed result map

8.1 Proposition 1: Communication equilibria

- Establishes the interval-partition form of equilibrium communication.
- Shows that the model supports equilibria with arbitrarily many intervals.
- Provides the foundation for treating communication quality continuously.

Interpretation: Communication can be partially informative rather than all-or-nothing. This makes it possible to compare vertical and horizontal communication intensities.

8.2 Proposition 2: Efficiency of the limiting equilibrium

- The limit as the number of intervals goes to infinity yields a perfect Bayesian equilibrium.
- This equilibrium maximizes expected profits among the communication equilibria considered.

Interpretation: The paper focuses on the most informative cheap-talk equilibrium, giving both organizational forms their best chance.

8.3 Lemma 1 and Proposition 3: Communication quality

- Lemma 1 derives residual variance as $S_l = b_l / (3 + 4b_l)$.
- Proposition 3 compares S_C and S_D across λ and δ .
- Vertical communication is generally better, but the gap vanishes when coordination becomes dominant.

Interpretation: These results drive the main theorem. They show that coordination needs alter communication incentives differently across organizational forms.

8.4 Proposition 4: Organizational performance

- Expected profits under both structures can be expressed as a function of uncertainty and residual variance.
- Centralization's loss is from information loss; decentralization's loss is from biased decision making and imperfect coordination.

Interpretation: This result decomposes the cost of each organizational form in a way that can be compared analytically.

8.5 Lemma 2 and Proposition 5: Centralization versus decentralization

- Lemma 2 proves that decentralization has lower adaptation losses and centralization has lower coordination losses.
- Proposition 5 states the conditions under which each structure dominates.
- Decentralization dominates for small own-division bias, for low coordination needs, and even for arbitrarily high coordination needs when bias is sufficiently small.

Interpretation: This is the paper's main result: centralization is not guaranteed by high coordination need. It requires high coordination need plus sufficiently high incentive conflict.

8.6 Extension results: sequential and asymmetric organizations

- Sequential decision making can favor centralization when coordination is very important.
- If the leader has more information, decentralization becomes more attractive.
- Differences in division size or coordination needs weaken but do not eliminate the core result.

Interpretation: The core mechanism is robust but context-dependent. The degree of symmetry affects the strength of the decentralization result.

9 Short summary

- The paper studies whether coordination inside organizations requires centralized decision making.
- The organization has two divisions, two local states, and two decisions.
- Managers privately observe local states and communicate strategically by cheap talk.
- Headquarters is unbiased but uninformed.
- Division managers are informed but biased toward their own divisions.
- Centralization improves coordination because one decision maker chooses both decisions.
- Decentralization improves adaptation because local managers directly use local information.
- Vertical communication is generally more informative than horizontal communication.
- A higher own-division bias worsens both forms of communication, especially horizontal communication.
- A higher need for coordination worsens vertical communication but improves horizontal communication.
- This means decentralization can dominate even when coordination is very important.

- Centralization is optimal only when coordination needs and own-division bias are both sufficiently large.
- The model provides empirical implications for firms facing coordination shocks and variation in division-manager compensation.

10 Conclusion

10.1 Core conclusion

Alonso, Dessein, and Matouschek show that coordination does not necessarily require centralization. Centralization coordinates decisions better, but it also relies on vertical communication from biased local managers to headquarters. When coordination becomes more important, vertical communication can deteriorate because managers have stronger incentives to influence the central compromise. By contrast, decentralized managers have stronger incentives to communicate accurately with each other because they directly bear the cost of coordination failure.

10.2 Economic intuition

The model's intuition can be summarized in one sentence: authority changes the sender's incentives to communicate. Under centralization, communication is persuasion directed at a central compromise. Under decentralization, communication is mutual adjustment among managers who retain local authority. The same coordination need that makes a central compromise valuable also makes local managers care more about each other's decisions.

10.3 Incremental contribution to organizational economics

The paper contributes a clean cheap-talk foundation for the delegation principle. It shows that the local-knowledge argument does not depend on physical communication costs. Even when communication is technologically free, strategic misrepresentation can make decentralized authority valuable. The contribution is especially important because it turns "coordination versus decentralization" from a binary debate into an endogenous communication-quality problem.

10.4 Implications for firm design

The model suggests that managers should not centralize simply because divisions need to coordinate. They should ask whether division managers have sufficiently balanced incentives. If division managers are rewarded mainly on divisional performance, centralization may be needed. If division managers are rewarded sufficiently on firm-wide performance, decentralized communication and authority may achieve coordination while preserving local adaptation.

10.5 Implications for empirical research

The paper points toward two empirical variables that should be measured jointly: the need for interdivisional coordination and the strength of own-division incentives. Organizational restructurings should be most likely when both variables are high. Coordination shocks should have stronger centralizing effects in firms where division managers have more divisional incentives.

10.6 Limitations

The model abstracts from many organizational features: repeated relationships, formal incentive contracts, multiple layers of hierarchy, effort provision, internal politics, and direct transfers. It also focuses on a symmetric two-division benchmark. These simplifications make the mechanism transparent but limit the direct mapping to complex firms.

10.7 Takeaway

The paper's lasting message is that organizational authority and organizational communication are inseparable. Decentralization can be an information-processing device, not merely an incentive device. Centralization is valuable when managers are biased and coordination is important, but decentralized authority can be superior when local knowledge is important and incentives are sufficiently aligned.